Python core programming

Units 1–5 · syllabus 2024, Calicut University BSc Physics Honours · module marks 12, doubled to 24

HOW THIS PAPER WORKS

The syllabus gives this module 12 marks. This paper gives **twice** that (24 marks), so every unit is tested with more questions than a real paper would use.

No past paper of this exact course was found, so the questions use similar CUFYUGP papers. **PYQ** = adapted from a similar paper, named with its code, term and question number. **Instructor** = written from the syllabus, with no close match. Each question also shows its unit (**U1** to **U5**). Every concept of units 1–5 is the main point of at least one question.

Similar papers used: CSC3MN206 Programming using Python, Nov 2025, versions A and B (**N25-A**, **N25-B**); MAT4CJ205 Fundamentals of Python and SageMath, Apr 2026 (**A26**).

Time 1 hour 30 minutes

Maximum 24 marks

Section A: answer all. Section B: answer any two of three. Write the Python code exactly as it would run in Jupyter or Colab.

Section A · Short answer

Answer all. Each question carries 2 marks. Total 16.

1. Name the three numerical data types in Python. Give one example of each, and state which one is written with a letter `j`.

PYQ N25-B Q3 · adapted · U1

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2. What is printed by the following code? Explain in one line why.

```
x = [1, 2, 3]
x.append(4)
print(x[1:3], len(x))
```

Instructor · U1

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3. State two differences between a list and a tuple. Why can a tuple not be changed after it is created?

PYQ N25-B Q18 · adapted · U2

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4. A student stores the pendulum period for each string length in a dictionary. Write one line of Python that creates the dictionary `T = {0.25: 1.00, 0.50: 1.42, 1.00: 2.01}` and one line that prints the period for length 0.50.

PYQ N25-A Q8 · adapted · U2

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5. Write the code that opens a text file named `data.txt` for reading, prints every line, and closes the file. Name the function used to open the file and the method used to close it.

Instructor · U2

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6. What does the following loop print? Explain the role of `break` in it.

```
for i in range(10):
    if i % 3 == 0:
        break
    print(i)
```

Instructor · U3

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7. Write a loop that prints the odd numbers from 1 to 9, skipping the number 5 with `continue`.

Instructor · U3

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8. Write one line of NumPy code that creates a 3 × 3 array of random integers from 1 to 10 (both included). Explain why the value written as 11 in the code is not included in the range.

PYQ A26 Q12 · adapted · U5

Section B · Descriptive

Answer any two of the three. Each carries 4 marks. Total 8.

9. Write a function `extremes(values)` that accepts a list of numbers and returns the largest and smallest numbers as a tuple. Explain what is returned, and state the output for `extremes([4, 9, -2, 7])`.

PYQ N25-A Q7 · adapted · U4

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10. Two 3 × 3 NumPy arrays `A` and `B` are created. Explain how `A + B`, `A - B` and `A * B` work element by element. Then explain what happens when `A / B` is computed and one element of `B` is zero.

PYQ N25-B Q6 · adapted · U5

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11. A student measures the period of a simple pendulum five times and stores the readings in a list `periods = [2.01, 1.98, 2.03, 2.00, 1.99]`. Write a function that computes the mean of the list using a loop and returns it, then print the mean rounded to two decimal places.

Instructor · U1, U4

COVERAGE

UNIT	TOPIC	QUESTIONS
1	Variables, operators, int / float / complex, lists	1, 2, 11
2	Tuples, sets, dictionary, input, open() and close()	3, 4, 5
3	if / else, break and continue	6, 7
4	Functions: arguments and return values	9, 11
5	NumPy arrays: creation, access, operations	8, 10

Every unit is covered. Total marks: 16 (Section A) + 8 (Section B) = 24.